

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	Speed (of drone) = $\frac{170}{32} = 5.3 \text{ [m s}^{-1}\text{]} (1)$ Displacement = 78.1 [m] (1) Velocity = $\frac{78.1(\text{ecf})}{32} = 2.4 \text{ [m s}^{-1}\text{]} (1)$ Ted correct (Bill incorrect) - only award mark for correct reasoning and consistent with calculated speed and velocity (1) Accept answers based on distance / displacement only i.e. Distance = 170 [m] (1) Displacement = 78.1 [m] (1) Time same for both stated (or equivalent) (1) Ted correct / Bill incorrect - only award mark if correct reasoning and consistent with calculated distance and displacement (1)			4	4	3	
		(d)	Benefits (1 from): • Surveying (general) or filming e.g archaeological sites, volcanoes, disaster areas etc • Monitoring (general) e.g wildlife • Potential for delivery of goods / Amazon • Aerial photography • Able to fly where other aircraft can't Risks (1 from): • Delivery of illegal goods e.g prison • Invasion of privacy • Air to air collisions • Disruption to society • Falling out of control • Risk of injury to humans and wildlife [due to the blades]			2	2		
			Question 3 total	5	6	6	17	9	0